

ABALAM
(Abacavir Sulfate and
Lamivudine Tablet
600mg / 300mg)
2079489

ABALAM

(Abacavir Sulfate and Lamivudine Tablet 600mg / 300mg)

NAME OF ACTIVE INGREDIENT

Each film-coated tablet contains Abacavir 600mg and Lamivudine 300mg.

DOSAGE FORM

Film coated tablet.

PRODUCT DESCRIPTION

The tablets are orange, modified capsule shaped, film coated tablets, debossed with ‘H’ on one side and ‘A1’ on the other side.

702 mg of Abacavir Sulfate is equivalent to 600 mg of Abacavir.

PHARMACODYNAMICS

Pharmacotherapeutic group: Antivirals for systemic use, antivirals for treatment of HIV infections, combinations. ATC code: J05AR02.

Mechanism of action: Abacavir and lamivudine are NRTIs and are potent selective inhibitors of HIV-1 and HIV-2 (LAV2 and EHO) replication. Both abacavir and lamivudine are metabolised sequentially by intracellular kinases to the respective 5'-triphosphate (TP) which are the active moieties. Lamivudine-TP and carbovir-TP (the active triphosphate form of abacavir) are substrates for and competitive inhibitors of HIV reverse transcriptase (RT). However, their main antiviral activity is through incorporation of the monophosphate form into the viral DNA chain, resulting in chain termination. Abacavir and lamivudine triphosphates show significantly less affinity for host cell DNA polymerases.

No antagonistic effects *in vitro* were seen with lamivudine and other antiretrovirals (tested agents: didanosine, nevirapine and zidovudine). The antiviral activity of abacavir in cell culture was not antagonized when combined with the nucleoside reverse transcriptase inhibitors (NRTIs) didanosine, emtricitabine, stavudine, tenofovir or zidovudine, the non-nucleoside reverse transcriptase inhibitor (NNRTI) nevirapine, or the protease inhibitor (PI) amprenavir.

Antiviral Activity in vitro

Both abacavir and lamivudine have been shown to inhibit replication of laboratory strains and clinical isolates of HIV in a number of cell types, including transformed T cell lines, monocyte/macrophage derived lines and primary cultures of activated peripheral blood lymphocytes (PBLs) and monocyte/macrophages. The concentration of drug necessary to effect viral replication by 50% (EC₅₀) or 50% inhibitory concentration (IC₅₀) varied according to virus and host cell type.

The mean EC₅₀ for abacavir against laboratory strains of HIV-1_{IIIB} and HIV-1_{HXB2} ranged from 1.4 to 5.8 µM. The median or mean EC₅₀ values for lamivudine against laboratory strains of HIV-1 ranged from 0.007 to 2.3 µM. The mean EC₅₀ against laboratory strains of HIV- 2 (LAV2 and EHO) ranged from 1.57 to 7.5 µM for abacavir and from 0.16 to 0.51 µM for lamivudine.

The EC₅₀ values of abacavir against HIV-1 Group M subtypes (A-G) ranged from 0.002 to 1.179 µM, against Group O from 0.022 to 1.21µM, and against HIV-2 isolates, from 0.024 to 0.49 µM. For lamivudine, the EC₅₀ values against HIV-1 subtypes (A-G) ranged from 0.001 to 0.170 µM, against Group O from 0.030 to 0.160 µM and against HIV-2 isolates from 0.002 to 0.120 µM in peripheral blood mononuclear cells.

Baseline HIV-1 samples from therapy-naïve subjects with no amino acid substitutions associated with *resistance* have been evaluated using either the multi-cycle Virco Antivirogram™ assay (n=92 from COL40263) or the the single cycle Monogram Biosciences PhenoSense™ assay (n=138 from ESS30009). These resulted in median EC₅₀ values of 0.912 µM (range: 0.493 to 5.017 µM) and 1.26 µM (range 0.72 to 1.91 µM) respectively for abacavir, and median EC₅₀ values of 0.429 µM (range: 0.200 to 2.007 µM) and 2.38 µM (1.37 to 3.68 µM) respectively for lamivudine.

Phenotypic susceptibility analyses of clinical isolates from antiretroviral-naïve patients with HIV-1 Group M non-B subtypes in three studies have each reported that all viruses were fully susceptible to both abacavir and lamivudine; one study of 104 isolates that included subtypes A and A1 (n=26), C (n=1), D (n=66), and the circulating recombinant forms (CRFs) AD (n=9), CD (n=1), and a complex inter-subtype recombinant_cpx (n=1), a second study of 18 isolates including subtype G (n=14) and CRF_AG (n=4) from Nigeria, and a third study of six isolates (n=4 CRF_AG, n=1 A and n=1 undetermined) from Abidjan (Côte d'Ivoire).

HIV-1 isolates (CRF01_AE, n=12; CRF02_AG, n=12; and Subtype C or CRF_AC, n=13) from 37 untreated patients in Africa and Asia were susceptible to abacavir (IC₅₀ fold changes <2.5), and lamivudine (IC₅₀ fold changes<3.0), except for two CRF02_AG isolates with fold-changes of 2.9 and 3.4 for abacavir. Group O isolates from antiviral naïve patients tested for lamivudine activity were highly sensitive.

The combination of abacavir and lamivudine has demonstrated antiviral activity in cell culture against non-subtype B isolates and HIV-2 isolates with equivalent antiviral activity as for subtype B isolates.

Resistance

In vivo resistance

Abacavir-resistant isolates of HIV-1 have been selected *in-vitro* in wild-type strain HIV-1 (HXB2) and are associated with specific genotypic changes in the RT codon region (codons M184V, K65R, L74V and Y115). Selection for the M184V mutation occurred first and resulted in a two-fold increase in IC₅₀. Continued passage in increasing concentrations of drug resulted in selection for double RT mutants 65R/184V and 74V/184V or triple RT mutant 74V/115Y/184V. Two mutations conferred a 7- to 8-fold change in abacavir susceptibility and combinations of three mutations were required to confer more than an 8-fold change in susceptibility. Passage with a zidovudine resistant clinical isolate RTMC also selected for the 184V mutation.

HIV-1 resistance to lamivudine involves the development of a M184I or, more commonly, M184V amino acid change close to the active site of the viral RT. Passage of HIV-1 (HXB2) in the presence of increasing 3TC concentrations results in high-level (>100 to >500- fold) lamivudine-resistant viruses and the RT M184I or V mutation is rapidly selected. The IC₅₀ for wild-type HXB2 is 0.24 to 0.6 µM, while the IC₅₀ for M184V containing HXB2 is >100 to 500 µM.

Antiviral therapy According to Genotypic/Phenotypic Resistance

In vivo resistance (Therapy-naïve patients)

The M184V or M184I variants arise in HIV-1 infected patients treated with lamivudine- containing antiretroviral therapy. Isolates from most patients experiencing virological failure with a regimen containing abacavir in pivotal clinical trials showed either no NRTI-related changes from baseline (45%) or only M184V or M184I selection (45%). The overall selection frequency for M184V or M184I was high (54%), and less common was the selection of L74V (5%), K65R (1%) and Y115F (1%). The inclusion of zidovudine in the regimen has been found to reduce the frequency of L74V and K65R selection in the presence of abacavir (with zidovudine: 0/40, without zidovudine: 15/192, 8%).

PHARMACOKINETICS

The fixed-dose combination tablet of abacavir/lamivudine (FDC) has been shown to be bioequivalent to lamivudine and abacavir administered separately. FDC can be taken with or without food. The pharmacokinetic properties of lamivudine and abacavir are described below.

Absorption

Abacavir and lamivudine are rapidly and well absorbed from the gastro-intestinal tract following oral administration. The absolute bioavailability of oral abacavir and lamivudine in adults is about 83% and 80-85% respectively. The mean time to maximal serum concentrations (t_{max}) is about 1.5 hours and 1.0 hour for abacavir and lamivudine, respectively. Following a single dose of 600 mg of abacavir, the mean (CV) C_{max} is 4.26 µg/ml (28%) and the mean (CV) AUC_∞ is 11.95 µg.h/ ml (21%). Following multiple-dose oral administration of lamivudine 300 mg once daily for seven days, the mean (CV) steady-state C_{max} is 2.04 µg/ml (26%) and the mean (CV) AUC₂₄ is 8.87 µg.h/ml (21%).

Distribution

Intravenous studies with abacavir and lamivudine showed that the mean apparent volume of distribution is 0.8 and 1.3 l/kg respectively. Plasma protein binding studies *in vitro* indicate that abacavir binds only low to moderately (~49%) to human plasma proteins at therapeutic concentrations. Lamivudine exhibits linear pharmacokinetics over the therapeutic dose range and displays limited plasma protein binding *in vitro* (< 36%). This indicates a low likelihood for interactions with other medicinal products through plasma protein binding displacement. Data show that abacavir and lamivudine penetrate the central nervous system (CNS) and reach the cerebrospinal fluid (CSF). Studies with abacavir demonstrate a CSF to plasma AUC ratio of between 30 to 44%. The observed values of the peak concentrations are 9-fold greater than the IC₅₀ of abacavir of 0.08 µg/ml or 0.26 µM when abacavir is given at 600 mg twice daily. The mean ratio of CSF/serum lamivudine concentrations 2-4 hours after oral administration was approximately 12%. The true extent of CNS penetration of lamivudine and its relationship with any clinical efficacy is unknown.

Biotransformation

Abacavir is primarily metabolised by the liver with approximately 2% of the administered dose being renally excreted, as unchanged compound. The primary pathways of metabolism in man are by alcohol dehydrogenase and by glucuronidation to produce the 5'- carboxylic acid and 5'- glucuronide which account for about 66% of the administered dose. These metabolites are excreted in the urine.

Metabolism of lamivudine is a minor route of elimination. Lamivudine is predominately cleared by renal excretion of unchanged lamivudine. The likelihood of metabolic drug interactions with lamivudine is low due to the small extent of hepatic metabolism (5-10%).

Elimination

The mean half-life of abacavir is about 1.5 hours. Following multiple oral doses of abacavir 300 mg twice a day there is no significant accumulation of abacavir. Elimination of abacavir is via hepatic metabolism with subsequent excretion of metabolites primarily in the urine. The metabolites and unchanged abacavir account for about 83% of the administered abacavir dose in the urine. The remainder is eliminated in the faeces.

The observed lamivudine half-life of elimination is 5 to 7 hours. The mean systemic clearance of lamivudine is approximately 0.32 l/h/kg, predominantly by renal clearance (> 70%) via the organic cationic transport system. Studies in patients with renal impairment show lamivudine elimination is affected by renal dysfunction. ABALAM is not recommended for use in patients with a creatinine clearance < 50 ml/min as necessary dose adjustment cannot be made.

Special patient populations

Hepatic impairment

Pharmacokinetic data has been obtained for abacavir and lamivudine separately.

Abacavir is metabolised primarily by the liver. The pharmacokinetics of abacavir have been studied in patients with mild hepatic impairment (Child-Pugh score 5-6) receiving a single 600 mg dose; the median (range) AUC value was 24.1 (10.4 to 54.8) ug.h/ml. The results showed that there was a mean (90%CI) increase of 1.89-fold [1.32; 2.70] in the abacavir AUC, and 1.58 [1.22; 2.04] fold in the elimination half-life. No definitive recommendation on dose reduction is possible in

patients with mild hepatic impairment due to substantial variability of abacavir exposure.

Data obtained in patients with moderate to severe hepatic impairment show that lamivudine pharmacokinetics are not significantly affected by hepatic dysfunction.

Based on data obtained for abacavir, ABALAM is not recommended in patients with moderate or severe hepatic impairment.

Renal impairment

Pharmacokinetic data have been obtained for lamivudine and abacavir alone. Abacavir is primarily metabolised by the liver with approximately 2% of abacavir excreted unchanged in the urine. The pharmacokinetics of abacavir in patients with end-stage renal disease is similar to (AUC) are increased in patients with renal dysfunction due to decreased clearance. ABALAM is not recommended for use in patients with a creatinine clearance < 50 ml/min as necessary dose adjustment cannot be made.

Elderly

No pharmacokinetic data are available in patients over 65 years of age.

Children

Abacavir is rapidly and well absorbed from oral formulations when administered to children. Paediatric pharmacokinetic studies have demonstrated that once daily dosing provides equivalent AUC₂₄ to twice daily dosing of the same total daily dose for both oral solution and tablet formulations.

The absolute bioavailability of lamivudine (approximately 58 to 66%) was lower and more variable in paediatric patients under 12 years of age. However, paediatric pharmacokinetic studies with tablet formulations have demonstrated that once daily dosing provides equivalent AUC₂₄ to twice daily dosing of the same total daily dose.

INDICATIONS

ABALAM is indicated in antiretroviral combination therapy for the treatment of Human Immunodeficiency Virus (HIV) infection in adults, adolescents and children weighing at least 25 kg.

Before initiating treatment with abacavir, screening for carriage of the HLA-B*5701 allele should be performed in any HIV-infected patient, irrespective of racial origin. Abacavir should not be used in patients known to carry the HLA-B*5701 allele.

RECOMMENDED DOSE

The recommended dose of ABALAM is one tablet once daily.

Children Under 25 kg:

ABALAM should not be administered to children who weigh less than 25 kg because it is a fixed-dose tablet that cannot be dose reduced.

ABALAM is a fixed-dose tablet and should not be prescribed for patients requiring dose adjustments. Separate preparations of abacavir or lamivudine are available in cases where discontinuation or dose adjustment of one of the active substances is indicated. In these cases, the physician should refer to the individual product information for these medicinal products.

Special Populations

Elderly: No pharmacokinetic data are currently available in patients over 65 years of age. Special care is advised in this age group due to age associated changes such as the decrease in renal function and alteration of haematological parameters.

Renal impairment:

ABALAM is not recommended for use in patients with a creatinine clearance < 50 ml/min as necessary dose adjustment cannot be made.

Hepatic impairment:

Abacavir is primarily metabolised by the liver. No clinical data are available in patients with moderate or severe hepatic impairment, therefore the use of ABALAM is not recommended unless judged necessary. In patients with mild hepatic impairment (Child-Pugh score 5-6) close monitoring is required, including monitoring of abacavir plasma levels if feasible.

Paediatric population:

The safety and efficacy of ABALAM in children weighing less than 25 kg has not been established. Currently available data are described but no recommendation on posology can be made.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

Hypersensitivity to the active substances or to any of the excipients.

WARNINGS AND PRECAUTIONS

The special warnings and precautions relevant to abacavir and lamivudine are included in this section. There are no additional precautions and warnings relevant to ABALAM.

While effective viral suppression with antiretroviral therapy has been proven to substantially reduce the risk of sexual transmission, a residual risk cannot be excluded. Precautions to prevent transmission should be taken in accordance with national guidelines.

Hypersensitivity Reactions

Abacavir is associated with a risk for hypersensitivity reactions (HSR) characterised by fever and/or rash with other symptoms indicating multi-organ involvement. HSRs have been observed with abacavir, some of which have been life-threatening, and in rare cases fatal, when not managed appropriately.

The risk for abacavir HSR to occur is high for patients who test positive for the HLA-B*5701 allele.

However, abacavir HSRs have been reported at a lower frequency in patients who do not carry this allele.

Therefore, the following should be adhered to:

- HLA-B*5701 status must always be documented prior to initiating therapy.
- ABALAM should never be initiated in patients with a positive HLA-B*5701 status, nor in patients with a negative HLA-B*5701 status who had a suspected abacavir HSR on a previous abacavir- containing regimen. (e.g. Ziagen)
- **ABALAM must be stopped without delay**, even in the absence of the HLA-B*5701 allele, if an HSR is suspected. Delay in stopping treatment with ABALAM after the onset of hypersensitivity may result in a life-threatening reaction.
- After stopping treatment with ABALAM for reasons of a suspected HSR, ABALAM **or any other medicinal product containing abacavir** (e.g. Ziagen) **must never be reinitiated**.
- Restarting abacavir containing products following a suspected abacavir HSR can result in a prompt return of symptoms within hours. This recurrence is usually more severe than on initial presentation and may include life-threatening hypotension and death.
- In order to avoid restarting abacavir, patients who have experienced a suspected HSR should be instructed to dispose of their remaining ABALAM tablets

Clinical Description of abacavir HSR

Abacavir HSR has been well characterised through clinical studies and during post marketing follow-up. Symptoms usually appeared within the first six weeks (median time to onset 11 days) of initiation of treatment with abacavir, **although these reactions may occur at any time during therapy**.

Almost all HSR to abacavir include fever and/or rash. Other signs and symptoms that have been observed as part of abacavir HSR are described in detail in section 4.8 (Description of selected adverse reactions), including respiratory and gastrointestinal symptoms. Importantly, such symptoms **may lead to misdiagnosis of HSR as respiratory disease (pneumonia, bronchitis, pharyngitis), or gastroenteritis**.

The symptoms related to HSR worsen with continued therapy and can be life-threatening. These symptoms usually resolve upon discontinuation of abacavir.

Rarely, patients who have stopped abacavir for reasons other than symptoms of HSR have also experienced life-threatening reactions within hours of re- initiating abacavir therapy. Restarting abacavir in such patients must be done in a setting where medical assistance is readily available.

Weight and metabolic parameters.

An increase in weight and in levels of blood lipids and glucose may occur during antiretroviral therapy. Such changes may in part be linked to disease control and life-style. For lipids, there is in some cases evidence for a treatment effect, while for weight gain there is no strong evidence relating this to any particular treatment. For monitoring of blood lipids and glucose reference is made to established HIV treatment guidelines. Lipid disorders should be managed as clinically appropriate.

Pancreatitis

Pancreatitis has been reported, but a causal relationship to lamivudine and abacavir is uncertain.

Risk of virological failure

- Triple nucleoside therapy: There have been reports of a high rate of virological failure, and of emergence of resistance at an early stage when abacavir and lamivudine were combined with tenofovir disoproxil fumarate as a once daily regimen.
- The risk of virological failure with ABALAM might be higher than with other therapeutic options.

Liver disease

The safety and efficacy of ABALAM has not been established in patients with significant underlying liver disorders. ABALAM is not recommended in patients with moderate or severe hepatic impairment.

Patients with pre-existing liver dysfunction, including chronic active hepatitis have an increased frequency of liver function abnormalities during combination antiretroviral therapy, and should be monitored according to standard practice. If there is evidence of worsening liver disease in such patients, interruption or discontinuation of treatment must be considered.

Patients co-infected with chronic hepatitis B or C virus

Patients with chronic hepatitis B or C and treated with combination antiretroviral therapy are at an increased risk of severe and potentially fatal hepatic adverse events. In case of concomitant antiviral therapy for hepatitis B or C, please refer also to the relevant product information for these medicinal products.

If lamivudine is being used concomitantly for the treatment of HIV and hepatitis B virus (HBV), additional information relating to the use of lamivudine in the treatment of hepatitis B infection can be found in the Summary of Product Characteristics for products containing lamivudine that are indicated for the treatment of HBV.

If ABALAM is discontinued in patients co-infected with HBV, periodic monitoring of both liver function tests and markers of

Size:240x390 mm

Pharma Code: Front: 6342 Back: 6343

No. of colors: 01, Black,

Note: Position, Height of the pharma code are tentative, it can be changed based on folding size.

HBV replication is recommended, as withdrawal of lamivudine may result in an acute exacerbation of hepatitis.

Mitochondrial dysfunction following exposure in utero.

Nucleoside and nucleotide analogues may impact mitochondrial function to a variable degree, which is most pronounced with stavudine, didanosine and zidovudine. There have been reports of mitochondrial dysfunction in HIV-negative infants exposed *in utero* and/or post- nately to nucleoside analogues: these have predominantly concerned treatment with regimens containing zidovudine. The main adverse reactions reported are haematological disorders (anaemia, neutropenia) and metabolic disorders (hyperlactatemia, hyperlipasemia). These reactions have often been transitory. Late onset neurological disorders have been reported rarely (hypertonia, convulsion, abnormal behaviour). Whether such neurological disorders are transient or permanent is currently unknown.

These findings should be considered for any child exposed *in utero* to nucleotide and nucleotide analogues, who presents with severe clinical findings of unknown etiology, particularly neurologic findings. These findings do not affect current national recommendations to use antiretroviral therapy in pregnant women to prevent vertical transmission of HIV.

Immune Reactivation Syndrome

In HIV-infected patients with severe immune deficiency at the time of institution of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic pathogens may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of CART. Relevant examples are cytomegalovirus retinitis, generalised and/or focal mycobacterial infections, and *Pneumocystis carinii* pneumonia. Any inflammatory symptoms should be evaluated and treatment instituted when necessary. Autoimmune disorders (such as Graves' disease) have also been reported to occur in the setting of immune reactivation; however, the reported time to onset is more variable and these events can occur many months after initiation of treatment.

Osteonecrosis

Although the etiology is considered to be multifactorial (including corticosteroid use, alcohol consumption, severe immunosuppression, higher body mass index), cases of osteonecrosis have been reported particularly in patients with advanced HIV-disease and/or long-term exposure to CART. Patients should be advised to seek medical advice if they experience joint aches and pain, joint stiffness or difficulty in movement.

Opportunistic infections

Patients should be advised that ABALAM or any other antiretroviral therapy does not cure HIV infection and that they may still develop opportunistic infections and other complications of HIV infection. Therefore, patients should remain under close clinical observation by physicians experienced in the treatment of these associated HIV diseases.

Myocardial infarction

Observational studies have shown an association between myocardial infarction and the use of abacavir. Those studied were mainly antiretroviral experienced patients. Data from clinical trials showed limited numbers of myocardial infarction and could not exclude a small increase in risk. Overall the available data from observational cohorts and from randomised trials show some inconsistency so can neither confirm nor refute a causal relationship between abacavir treatment and the risk of myocardial infarction. To date, there is no established biological mechanism to explain a potential increase in risk. When prescribing ABALAM, action should be taken to try to minimize all modifiable risk factors (e.g. smoking, hypertension, and hyperlipidaemia).

Drug Interactions:

ABALAM should not be taken with any other medicinal products containing lamivudine or medicinal products containing emtricitabine.

The combination of lamivudine with cladribine is not recommended.

Excipients

ABALAM contains the azo colouring agent sunset yellow, which may cause allergic reactions.

INTERACTIONS WITH OTHER MEDICAMENTS

ABALAM contains abacavir and lamivudine, therefore any interactions identified for these individually are relevant to ABALAM. Clinical studies have shown that there are no clinically significant interactions between abacavir and lamivudine.

Abacavir is metabolised by UDP-glucuronyltransferase (UGT) enzymes and alcohol dehydrogenase; co-administration of inducers or inhibitors of UGT enzymes or with compounds eliminated through alcohol dehydrogenase could alter abacavir exposure. Lamivudine is cleared renally. Active renal secretion of lamivudine in the urine is mediated through organic cation transporters (OCTs); co- administration of lamivudine with OCT inhibitors may increase lamivudine exposure.

Abacavir and lamivudine are not significantly metabolised by cytochrome P450 enzymes (such as CYP 3A4, CYP 2C9 or CYP 2D6) nor do they inhibit or induce this enzyme system. Therefore, there is little potential for interactions with antiretroviral protease inhibitors, non- nucleosides and other medicinal products metabolised by major P450 enzymes.

ABALAM should not be taken with any other medicinal products containing lamivudine.

Paediatric population

Interaction studies have only been performed in adults.

PREGNANCY AND LACTATION

Pregnancy

As a general rule, when deciding to use antiretroviral agents for the treatment of HIV infection in pregnant women and consequently for reducing the risk of HIV vertical transmission to the newborn, the animal data as well as the clinical experience in pregnant women should be taken into account.

Animal studies with abacavir have shown toxicity to the developing embryo and foetus in rats, but not in rabbits. Animal studies with lamivudine showed an increase in early embryonic deaths in rabbits but not in rats. The active ingredients of abacavir/lamivudine may inhibit cellular DNA replication and abacavir has been shown to be carcinogenic in animal models. The clinical relevance of these findings is unknown. Placental transfer of abacavir and lamivudine has been shown to occur in humans.

In pregnant women treated with abacavir, more than 800 outcomes after first trimester exposure and more than 1000 outcomes after second and third trimester exposure indicate no malformative and foeto/neonatal effect. In pregnant women treated with lamivudine, more than 1000 outcomes from first trimester and more than 1000 outcomes from second and third trimester exposure indicate no malformative and foeto/neonatal effect. There are no data on the use of abacavir/ lamivudine in pregnancy, however the malformative risk is unlikely in humans based on those data.

For patients co-infected with hepatitis who are being treated with a lamivudine containing medicinal product such as Abacavir/Lamivudine and subsequently become pregnant, consideration should be given to the possibility of a recurrence of hepatitis on discontinuation of lamivudine.

Mitochondrial dysfunction

Nucleoside and nucleotide analogues have been demonstrated *in vitro* and *in vivo* to cause a variable degree of mitochondrial damage. There have been reports of mitochondrial dysfunction in HIV-negative infants exposed *in utero* and/or post-natally to nucleoside analogues.

Breast-feeding

Abacavir and its metabolites are excreted into the milk of lactating rats. Abacavir is also excreted into human milk.

Based on more than 200 mother/child pairs treated for HIV, serum concentrations of lamivudine in breastfed infants of mothers treated for HIV are very low (< 4% of maternal serum concentrations) and progressively decrease to undetectable levels when breastfed infants reach 24 weeks of age. There are no data available on the safety of abacavir and lamivudine when administered to babies less than three months old.

It is recommended that HIV infected women do not breast-feed their infants under any circumstances in order to avoid transmission of HIV.

Fertility

Studies in animals showed that neither abacavir nor lamivudine had any effect on fertility.

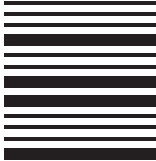
SIDE EFFECTS

Summary of the safety profile

The adverse reactions reported for abacavir/lamivudine were consistent with the known safety profiles of abacavir and lamivudine when given as separate medicinal products. For many of these adverse reactions it is unclear whether they are related to the active substance, the wide range of other medicinal products used in the management of HIV infection, or whether they are a result of the underlying disease process.

Many of the adverse reactions listed in the table below occur commonly (nausea, vomiting, diarrhoea, fever, lethargy, rash) in patients with abacavir hypersensitivity. Therefore, patients with any of these symptoms should be carefully evaluated for the presence of this hypersensitivity. Very rarely cases of erythema multiforme, Stevens-Johnson syndrome or toxic epidermal necrolysis have been reported where abacavir hypersensitivity could not be ruled out. In such cases medicinal products containing abacavir should be permanently discontinued.

Body System	Abacavir	Lamivudine
Blood and lymphatic systems disorders		Uncommon:Neutropenia and anaemia (both occasionally severe), thrombocytopenia Very rare: Pure red cell aplasia
Immune system disorders	Common: hypersensitivity	
Metabolism and nutrition disorders	Common: anorexia Very rare: lactic acidosis	Very rare: lactic acidosis
Nervous system disorders	Common: headache	Common: headache, insomnia Very rare: cases of peripheral neuropathy (or paraesthesia) havebeen reported
Respiratory, thoracic and mediastinal disorders		Common: cough, nasal symptoms



Gastrointestinal disorders	Common: nausea, vomiting, diarrhoea Rare: pancreatitis have been reported, but a casual relationship to abacavir treatment is uncertain	Common: nausea, vomiting, abdominal pain or cramps, diarrhoea Rare: rises in serum amylase. Cases of pancreatitis have been reported.
Hepatobiliary disorders		Uncommon: transient rises in liver enzymes (AST, ALT) Rare: hepatitis
Skin and subcutaneous tissue disorders	Common: rash (without systemic symptoms) Very rare: erythema multiforme, Stevens-Johnson syndrome and toxic epidermal necrolysis	Common: rash, alopecia Rare: angioedema
Musculoskeletal and connective tissue disorders		Common: arthralgia, muscle disorders Rare: rhabdomyolysis
General disorders and administration site conditions	Common: fever, lethargy, fatigue	Common: fatigue, malaise, fever

Description of selected adverse reactions

Abacavir hypersensitivity

The signs and symptoms of this HSR are listed below. These have been identified either from clinical studies or post marketing surveillance. Those reported in at least 10% of patients with a hypersensitivity reaction are in bold text:

Almost all patients developing hypersensitivity reactions will have fever and/or rash (usually maculopapular or urticarial) as part of the syndrome, however, reactions have occurred without rash or fever. Other key symptoms include gastrointestinal, respiratory or constitutional symptoms such as lethargy and malaise.

Skin: Rash (usually maculopapular or urticarial)

Gastrointestinal tract: Nausea, vomiting, diarrhoea, abdominal pain, mouth ulceration

Respiratory tract: Dyspnoea, cough, sore throat, adult respiratory distress syndrome, respiratory failure.

Miscellaneous: Fever, lethargy, malaise, oedema, lymphadenopathy, hypotension, conjunctivitis, anaphylaxis

Neurological/Psychiatry: Headache, paraesthesia.

Haematological: Lymphopenia.

Liver / pancreas: Elevated liver function tests, hepatitis, hepatic failure.

Musculoskeletal: Myalgia, rarely myolysis, arthralgia, elevated creatine phosphokinase;

Urology: Elevated creatinine, renal failure.

Symptoms related to this HSR worsen with continued therapy and can be life-threatening and in rare instance, have been fatal.

Restarting abacavir following an abacavir HSR results in a prompt return of symptoms within hours. This recurrence of the HSR is usually more severe than on initial presentation and may include life-threatening hypotension and death. Similar reactions have also occurred infrequently after restarting abacavir in patients who had only one of the key symptoms of hypersensitivity (see above) prior to stopping abacavir; and on very rare occasions have also been seen in patients who have restarted therapy with no preceding symptoms of a HSR (i.e., patients previously considered to be abacavir tolerant).

Metabolic parameters

Weight and level of blood lipids and glucose may increase during antiretroviral therapy.

Immune reactivation syndrome

In HIV-infected patients with severe immune deficiency at the time of initiation of combination antiretroviral therapy, an inflammatory reaction to asymptomatic or residual opportunistic infections may arise. Autoimmune disorders (such as Graves' disease) have also been reported to occur in the setting of immune reconstitution; however, the reported time to onset is more variable and these events can occur many months after initiation of treatment.

Osteonecrosis

Cases of osteonecrosis have been reported, particularly in patients with generally acknowledged risk factors, advance HIV disease or long-term exposure to CART. The frequency of this is unknown.

Paediatric population

The safety database support once daily dosing in paediatric patients.

SYMPTOMS AND TREATMENT OF OVERDOSE

No specific symptoms or signs have been identified following acute overdose with abacavir or lamivudine, apart from those listed as undesirable effects.

If overdose occurs the patient should be monitored for evidence of toxicity, and standard supportive treatment applied as necessary. Since lamivudine is dialysable, continuous haemodialysis could be used in the treatment of overdose, although this has not been studied. It is not known whether abacavir can be removed by peritoneal dialysis or haemodialysis.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

No studies on the effects on ability to drive and use machines have been performed. The clinical status of the patient and the adverse reaction profile of abacavir/lamivudine should be borne in mind when considering the patient's ability to drive or operate machinery.

PHARMACEUTICAL PARTICULARS

List of excipients:

Tablet Core: Microcrystalline Cellulose, Sodium Starch Glycolate, Colloidal Silicon Dioxide, Magnesium Stearate, Purified Water

Tablet film coat: Hypromellose (E464), Titanium Dioxide (E171), Macrogol (E1521), Sunset Yellow FCF Aluminium Lake (E110), Polysorbate 80 (E433).

Storage Condition

Store below 30°C and protect from moisture.

Packaging Available: Pack of 30 tablets in Alu-Alu blister

MANUFACTURED BY:

Hetero Labs Limited

Unit V, Survey No. 439, 440, 441 & 458,
TSIIC Formulation SEZ, Jadcherla Mandal,
Mahaboob nagar (Dist.) - 509301,
Telangana, India.

PRODUCT REGISTRATION HOLDER:

CAMBER LABORATORIES SDN BHD
Unit E-13A-02, Menara SUEZCAP 2, No. 2, KL Gateway,
Jalan Kerinchi, Gerbang Kerinchi Lestari,
59200 Kuala Lumpur, Malaysia.

Date of Revision: October 2024